

Tofa Kimani Mwangi

Data Engineer | Software Engineering & Big Data

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Professional Summary

Data Engineer & Full-Stack Developer with a background in Applied Computing, bridging software engineering, live data analysis, and innovation-driven research. Skilled in building robust ETL pipelines, full-stack web applications (Next.js) with databases such as PostgreSQL, AI prompting, and translating real-time insights into actionable strategy. Passionate about optimizing data workflows and leveraging modern tech to deliver scalable, high-impact digital solutions

Skills

Software Engineer

MySQL (SQL), Python, Node.js, Express, Bootstrap, APIs Databases: MySQL, MongoDB Version Control & Tools: Git, GitHub, Javascript, Tailwind CSS, Java, RESTful APIs, Springboot

Data Engineering

Data Cleaning, Big Data Processing, NumPy, Python, Matplotlib, Pandas, ETL Pipelines, Data Architect, Data Modelling, Relational Modeling (RDBMS), Data Warehousing Concepts, Data Architecture, Docker

Tools & Office Suite

Jupyter Notebook, Anaconda, Excel(Advanced), MySQL, Git, Github, PostgreSQL, CI/CD basics, Big Query, Docker, Microsoft PowerPoint, Microsoft Word

Professional Skills

Project Development, Documentation, Analytical Thinking, Technical Troubleshooting, System Optimization, Database Management, Debugging, Automation, Prompt Engineering, Technical Support, Agile/Scrum Methodology, Test-Driven Development

Skills

Teamwork, Collaboration, Communication, Presentation, Leadership, Decision-Making, Problem-Solving, Critical Thinking, Time Management, Adaptability

Innovation Skills

Data Storytelling, Business Analytics, Research, Cross-Functional Collaboration., Trend & Pattern Identification, User/Customer Insights

Experience

Social Health Authority Nanyuki

IT Intern September 2024 - December 2024

Optimized data integrity by spearheading the cleaning and preparation of large-scale datasets using Excel and internal diagnostic tools, reducing reporting errors.

Resolved complex technical issues across computer systems and networks, maintaining high uptime for critical user applications.

Collaborated with cross-functional teams to redesign IT workflows, significantly enhancing the efficiency of data entry and report generation processes.

Conducted operational data analysis to identify workflow bottlenecks, collaborating with cross-functional teams to redesign internal processes and improve reporting efficiency.

Minerva Analytics Remote

Corporate Governance Data Analyst Jan 2026 – Apr 2026

• **Governance & Board Analytics:** Reviewed public company filings, annual reports, and proxy statements to audit, extract, and log board structure changes, executive leadership shifts, and committee appointments.

• **Financial Data Validation:** Captured and verified financial metrics, director independence data, and board governance updates with high accuracy across global markets.

Projects

1) Chap Mall May 2025 - July 2025

• **Full-Stack Architecture:** Architected and deployed a multi-tier e-commerce platform featuring role-based access control (RBAC) for customers, vendors, and administrators.

• **API & Data Integrity:** Engineered RESTful APIs and optimized database queries (PostgreSQL/SQL) to handle complex relational data, ensuring 100% data integrity for orders and inventory.

• **Real-Time Logic:** Built high-performance inventory tracking logic to power live availability updates, order processing, and streamlined dashboard reporting.

• **Security & Performance:** Implemented secure authentication protocols and optimized query execution to significantly reduce page latency and protect vendor/customer data.

<https://github.com/TofaKM/Advanced-E-commerce-Website->

2) Maji Ndogo - ALX December 2025

Agricultural Data Analytics Challenge

• Spearheaded deep-dive **data research** on agricultural datasets using Python to model crop yield variables and environmental impact metrics.

• Utilized **statistical libraries** to run live analyses, transforming messy raw data into clean, structured insights.

• Built visual **reports** bridging technical data findings with high-level recommendations for sustainable resource management.

3) Hospital ETL Pipeline 17 June 2026

• Built and maintained a **containerized (Docker) ETL pipeline** that streamlined healthcare data ingestion, cleaning, and transformation processes.

• Structured **end-to-end data workflows** to generate high-precision datasets optimized for live operational analysis and reporting.

• **Automated routine** transformations, enabling non-technical stakeholders to access reliable business insights and clinical reporting effortlessly.

https://github.com/TofaKM/etl_hospital_pipeline

4) Zuri Mobile Customer Churn ETL Pipeline June 2026

• Built a production-style telecom churn pipeline processing **10,000 records** using **PySpark** for validation, deduplication, cleaning, standardization, and domain checks.

• Designed a **PostgreSQL star-schema data warehouse** for efficient analytics and reporting.

• Orchestrated **reliable, repeatable staging-to-live data loads** using **Apache Airflow** and **Docker Compose**.

<https://github.com/TofaKM/zuri-mobile-churn-pipeline>

Certifications

Data Engineer March 2025 - August 2026

ALX Africa

• Mastered **ETL/ELT pipeline development**, data integration, and transformation workflows.

• Developed strong skills in **Data Warehousing and dimensional data modeling**, including fact and dimension tables.

• Built and orchestrated data pipelines using **Apache Airflow**.

• Worked with **Apache Spark** for distributed data processing and large-scale data transformation.

• Applied **Python and SQL** for data extraction, transformation, analysis, and automation.

• Gained practical knowledge of **data engineering security**, including access control, data protection, and secure pipeline practices.

• Worked with modern data engineering concepts and tools to design **reliable, scalable, and maintainable data pipelines**.

<https://ehub.alxafrica.com/ob3/verify-certificate/aac32928-744d-42d4-beff-85b393b73d1a>

Data Analytics 26 September 2025

ALX Africa

Specialized in **MySQL** for database querying, **Power BI** for dashboard creation, and professional data reporting

<https://savanna.alxafrica.com/certificates/97rGSZN3hJ>

Python 17 April 2026

ALX Africa

Leveraged **Pandas, NumPy**, and **SciPy** to engineer automated **ETL pipelines** and perform **statistical hypothesis testing-**

, transforming millions of messy records into clean datasets and actionable visual insights

<https://savanna.alxafrica.com/certificates/n8CmfHBhPR>

Data Analytics August 2024

Cisco Academy

Focused on the end-to-end data lifecycle: **cleaning, analysis, and visualization** using Excel and SQL.

https://www.credly.com/badges/aa459c6a-f252-45e4-b56a-49f9cd516354/public_url

Introduction to Machine Learning February 2025

Cisco IBM

Gained practical exposure to **Supervised and Unsupervised Learning** models and model evaluation using Python

https://www.credly.com/badges/d380bf99-8288-47f9-9883-c848a62d8460/public_url

Education

1) KCA University Bachelor of Science in Applied Computing

Computer Science Nairobi • September 2020 - November 2025

Academic Experience

Mastered Programming Concepts across multiple paradigms, specializing in object-oriented programming (OOP) and functional programming to write clean, maintainable code.

Deepened technical expertise in Data Structures and Algorithms (DSA), mastering the selection and implementation of optimal data types and algorithms to maximize software performance and memory efficiency.

Studied Systems Analysis & Design, with a focus on high-level architectural modeling, scalability strategies, and the end-to-end **Software Development Life Cycle (SDLC)** to build robust enterprise applications.

Advanced proficiency in Database Management Systems (DBMS), utilizing entity-relationship diagrams (ERDs) and normalization techniques to ensure data integrity and query efficiency across complex relational databases.

Final Year Project: Engineered a full-stack E-commerce platform leveraging **JavaScript, Node.js, React, and MySQL**, integrating system design principles to manage complex relational data and role-based access control.

2) Ndururumo High School Nyahururu • January 2016 - November 2019

Achieved a Mean Grade of B+ (71 points), maintaining a high standard of academic excellence in a rigorous curriculum focused on Mathematics, Physics, and Computer Studies.

Served as Academic Secretary in the final year, acting as the primary liaison between the student body and faculty to coordinate academic schedules and peer-to-peer tutoring initiatives.

Demonstrated strong quantitative aptitude by representing the school in regional Mathematics Competitions, finishing among the top participants.

Built a technical foundation in Computer Studies, gaining early exposure to systems logic and digital literacy that informed a career path in Software Engineering.

Profiles

GitHub **LinkedIn**

[Tofa_KM](#) [Tofa Mwangi](#)

Interests

Interests
Machine Learning, Analytics, Mobile App Development, Open Source, Tech Communities, Video Editing, Digital Storytelling, Badminton, Outdoor Activities, Football, Youtube, AI & Data Science, System Architecture

References

Faith Muna **Dacosta Otekoh** **Francis Ngugi**

Claim Officer / Supervisor Human Resources (HR) Software Engineer

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